

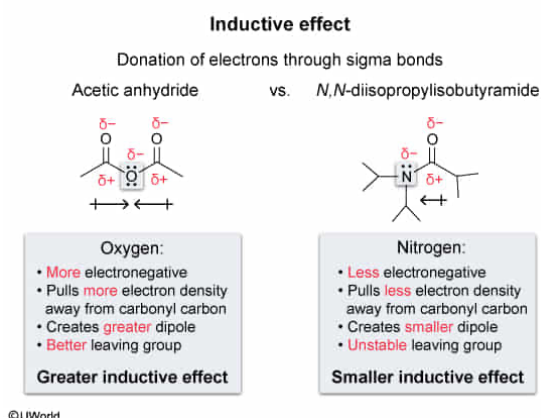
Which of the following does NOT describe why a nucleophile will react more quickly with acetic anhydride than with *N,N*-diisopropylisobutyramide?

- ✔ ☒ A. *N,N*-diisopropylisobutyramide has a greater inductive effect than acetic anhydride. [61%]
- ☐ B. Acetic anhydride is a stronger electrophile than *N,N*-diisopropylisobutyramide. [8%]
- ☐ C. Acetic anhydride is less sterically hindered than *N,N*-diisopropylisobutyramide. [9%]
- ☐ D. *N,N*-diisopropylisobutyramide has a less stable leaving group than acetic anhydride. [20%]

Correct

61% answered correctly

Explanation:



Reactivity of **carboxylic acid derivatives** with a **nucleophile** depends on various factors, including inductive effects, steric effects, and leaving group stability.

Induction is an electronic property in which **electrons** are **donated** through **sigma bonds**. **Electronegative** atoms or **electron withdrawing substituents** pull electrons away from an adjacent atom toward themselves, creating a **dipole**. The closer the electron withdrawing group to an atom, the greater the inductive effect that atom experiences.

Acetic anhydride and *N,N*-diisopropylisobutyramide both have two electronegative atoms bonded to each carbonyl carbon. The anhydride carbonyl carbons are bound to two oxygen atoms whereas the amide carbonyl carbon is bound to an oxygen and a nitrogen. Because oxygen is more electronegative than nitrogen, it draws more electron density away from the carbonyl carbon than nitrogen does, creating a larger dipole on the anhydride than on the amide, as well as a more electrophilic carbonyl carbon on the anhydride (**Choice B**).

Electron withdrawing groups also make **good leaving groups**. The leaving group of *N,N*-diisopropylisobutyramide is the amino group, and the leaving group of acetic anhydride is the carboxyl group. Amine groups are less stable leaving groups than carboxyl groups (**Choice D**) because carboxyl groups are stabilized by **resonance** delocalization of the negative charge obtained after being eliminated.

Nucleophiles react more readily with **less sterically hindered electrophiles**. The methyl groups bonded to the carbonyls on acetic anhydride provide less steric hindrance than the bulky

isopropyl groups on the nitrogen and adjacent to the carbonyl in *N,N*-diisopropylisobutyramide (**Choice C**), making it easier for a nucleophile to react with acetic anhydride.

The question asks which statement does NOT describe why a nucleophile will react more quickly with acetic anhydride than with *N,N*-diisopropylisobutyramide. Because nucleophiles react more readily with stronger electrophiles and anhydrides contain stronger electron withdrawing groups than amides, **anhydrides** experience a **greater inductive effect** and **react** with a nucleophile **more readily** than **amides**.

Therefore, *N,N*-diisopropylisobutyramide has a *smaller* rather than a greater inductive effect than acetic anhydride.

Educational objective:

Reactivity of carboxylic acid derivatives depends on various factors, including inductive effects, steric effects, and leaving group stability. The inductive effect is an electronic property in which the electrons are donated through sigma bonds. Anhydrides contain stronger electron withdrawing groups than amides; therefore, anhydrides experience a greater inductive effect and react more readily than amides.

Time spent:0 sec

Subject:
Organic Chemistry

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System:
Functional Groups and Their Reactions